

Temperature-Based Fan-Speed Controller

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The circuit presented here is an automatic fan-speed controller that automatically adjusts the fan's speed according to ambient temperature. It can be used in cooling electronic devices, where the fan speed

PARTS LIST

Semiconductors:

- IC1 - LM35 temperature sensor
- IC2 - LM358 operational amplifier
- IC3 - MCT2E optocoupler
- IC4 - NE555 timer
- IC5 - MOC3061 zero-cross optocoupler
- D1-D6 - 1N4007 rectifier diode
- TRIAC1 - BT136 TRIAC

Resistors (all $\frac{1}{4}$ -watt, $\pm 5\%$ carbon):

- R1, R3, R5 - 10 kilo-ohm
- R2 - 150 kilo-ohm
- R4 - 4.7 kilo-ohm
- R6 - 47 ohm
- R7 - 100 kilo-ohm
- R8, R9 - 330 ohm
- R10 - 220 ohm
- VR1 - 100 kilo-ohm preset

Capacitors:

- C1 - 470 μ F, 35V electrolytic
- C2 - 0.1 μ F ceramic disk
- C3 - 0.01 μ F, 220V polyester

Miscellaneous:

- CON1, CON2 - 2-pin connector
- X1 - 230V AC primary to 12V-0-12V, 300mA secondary transformer

needs to increase if heat dissipation increases.

Circuit and working

The circuit is built around temperature sensor LM35 (IC1), operational amplifier LM358 (IC2), opto-coupler MCT2E (IC3), timer NE555 (IC4), zero-cross photo-triac driver opto-coupler MOC3061 (IC5) and triac BT136 (TRIAC1), along with some other easily available components.

Temperature sensor IC1 senses changes in temperature and its output voltage varies linearly with the variation in temperature. The output

Test Points

Test point	Details
TP0	0V, GND
TP1	Around 16V
TP2	Trigger pulses for IC4 (around 100Hz)
TP3	Output of IC4 (pulse width depends on the level at pin 5 of IC4)
TP4	Amplified voltage corresponding to temperature

voltage of IC1 is amplified using operational amplifier IC2. This amplified voltage is then applied to control voltage pin 5 of IC4, which is wired as a monostable multivibrator.

The control voltage adjusts the threshold and triggering levels of timer IC4 and, hence, there are changes in the pulse width of the output waveform corresponding to the voltage level at pin 5 of IC4. The trigger pulses for the monostable multivibrator are produced through IC3. Opto-coupler IC3 is wired as a zero-crossing detector. As a result, a pulse-width-modulated waveform can be obtained in the output of the monostable multivibrator.

The output pulses of IC4 at pin 3 are applied to anode pin 1 of IC5, which is an optically isolated triac driver with zero-crossing detection. The zero-crossing facility in the triac driver ensures that the firing pulses are applied at exact zero crossing of the AC waveform.

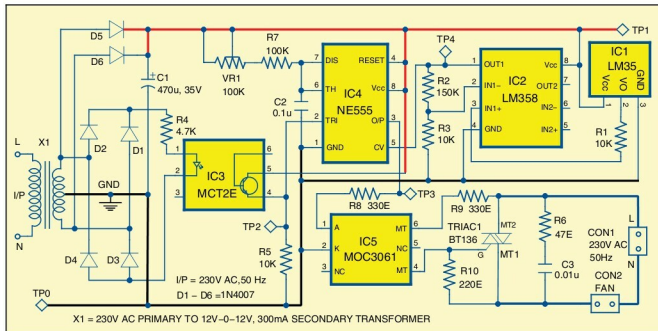


Fig. 1: Circuit of temperature-based fan-speed controller